

# Carter K. Cocke

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391 S Holliston Ave, MC 104-44, Pasadena, CA 91125

## Education

### California Institute of Technology

Expected 2027

*Ph.D. in Mechanical Engineering, GPA: 4.0*

Pasadena, CA

- Advisor: Prof. Kaushik Bhattacharya

### University of Utah

2022

*M.S. in Mechanical Engineering, GPA: 4.0*

Salt Lake City, UT

- Advisor: Prof. Ashley Spear

### University of Utah

2022

*Honors B.S. in Mechanical Engineering, magna cum laude, GPA: 3.98*

Salt Lake City, UT

- Solid Mechanics Emphasis, Honors Ecology and Legacy Minor

## Research Experience

### California Institute of Technology

Sept. 2022 – Current

*Graduate Research Assistant*

Pasadena, CA

Funded by: National Science Foundation Graduate Research Fellowship (NSF GRFP)

- Built a gradient-based shared-memory parallel optimization framework from scratch in C++ to recover full-field strains from raw high-energy 3D X-ray diffraction data, including detector data processing and handling, forward diffraction modeling, and PDE-constrained optimization
- Developed and implemented an MPI-parallelized finite-element-based (deal.II) iterative least-squares solver to post-process data recovered from high-energy X-ray diffraction experiments

### University of Utah

Aug. 2019 – Aug. 2022

*Graduate Research Assistant*

Salt Lake City, UT

Funded by: Department of Energy (DOE) Nuclear Energy University Program (NEUP)

Thesis title: Implementation and experimental validation of nonlocal damage in a large-strain elasto-viscoplastic FFT-based framework for predicting ductile fracture in 3D polycrystalline materials

- Implemented and validated a novel ductile fracture-extended finite-strain FFT-based crystal plasticity formulation and published the work in the *International Journal of Plasticity*
- Led and organized team efforts in a winning blind submission of simulation predictions to the NIST AM-Bench Challenge using the aforementioned fracture-extended FFT-based formulation
- Collaborated with three leading mechanics and materials science researchers to win the 2022 AFRL Challenge #4. Published a journal article in *Integrating Materials Manufacturing and Innovation* detailing the modeling approach and in-depth post-challenge investigation
- Developed serial and shared/distributed memory parallel Python, Fortran, and C++ code for computational solid mechanics and materials science research applications
- Leveraged high-performance computing (HPC) for running OpenMP and MPI parallelized codes on a Linux cluster environment using the Slurm workload manager

### Los Alamos National Laboratory

May 2021 – Aug. 2021

*Graduate Research Assistant*

Remote

- Developed a method to model mechanical indentation using a finite-strain FFT-based model through formulation modifications and novel algorithmic implementations
- Modernized and parallelized a Fortran77 crystal plasticity code via serial code optimizations, OpenMP parallelization, and external library implementations

## Publications

3. **C. K. Cocke**, A. Akerson, S. Gorske, K. Faber, K. Bhattacharya, “On recovering intragranular strain fields from grain-averaged strains obtained by high-energy X-ray diffraction microscopy,” *Mathematics and Mechanics of Solids*, Special Issue for Rohan Abeyaratne, pp. 1–18, 2026. <https://doi.org/10.1177/10812865251376693>
2. **C. K. Cocke**, H. Mirmohammad, M. Zecevic, B. Phung, R. Lebensohn, O. Kingstedt, A. Spear, “Implementation and experimental validation of nonlocal damage in a large-strain elasto-viscoplastic FFT-based framework for predicting ductile fracture in 3D polycrystalline materials,” *International Journal of Plasticity*, vol. 162, p. 103508, 2023. <https://doi.org/10.1016/j.ijplas.2022.103508>
1. **C. K. Cocke**, A. Rollett, R. Lebensohn, A. Spear, “The AFRL Additive Manufacturing Modeling Challenge: Predicting micromechanical fields in AM IN625 using an FFT-based method with direct input from a 3D microstructural image,” *Integrating Materials Manufacturing and Innovation*, vol. 10, no. 2, pp. 157–176, 2021. <https://doi.org/10.1007/s40192-021-00211-w>

## Conference Presentations

\* Indicates presenter

9. **C. K. Cocke\***, S. Gorske, K. Faber, K. Bhattacharya, “A gradient-based approach for recovering admissible intragranular strain and orientation fields from high-energy X-ray diffraction data,” *TMS 2026*, San Diego, CA, March 2026
8. **C. K. Cocke\***, A. Akerson, S. Gorske, K. Faber, K. Bhattacharya, “On the inverse problem of recovering admissible intragranular strain fields from high-energy X-ray diffraction data,” *3DMS 7*, Anaheim, CA, June 2025
7. **C. K. Cocke\***, A. Akerson, S. Gorske, K. Faber, K. Bhattacharya, “On the inverse problem of recovering admissible intragranular strain fields from high-energy X-ray diffraction data,” *TMS 2025*, Las Vegas, NV, March 2025
6. **C. K. Cocke\***, A. Akerson, S. Gorske, K. Faber, K. Bhattacharya, “On recovering intragranular strain fields from grain-averaged strains obtained by high-energy X-ray diffraction microscopy,” *Microstructure-Aware Modeling of the Mechanics of Heterogeneous Materials: a symposium honoring Ricardo Lebensohn*, Santa Fe, NM, April 2024
5. **C. K. Cocke\***, H. Mirmohammad, M. Zecevic, B. Phung, R. Lebensohn, O. Kingstedt, A. Spear, “Implementation and experimental validation of nonlocal damage in a large-strain elasto-viscoplastic FFT-based framework for predicting ductile fracture in 3D polycrystalline materials,” *WCCM 15*, Virtual, August 2022
4. **C. K. Cocke\***, H. Mirmohammad, M. Zecevic, B. Phung, R. Lebensohn, O. Kingstedt, A. Spear, “Implementation and experimental validation of nonlocal damage in a large-strain elasto-viscoplastic FFT-based framework for predicting ductile fracture in 3D polycrystalline materials,” *ESMC 11*, Galway, Ireland, July 2022
3. **C. K. Cocke\***, H. Mirmohammad, M. Zecevic, B. Phung, R. Lebensohn, O. Kingstedt, A. Spear, “Implementation and experimental validation of nonlocal damage in a large-strain elasto-viscoplastic FFT-based framework for predicting ductile fracture in 3D polycrystalline materials,” *3DMS 6*, Washington, D.C., June 2022
2. **C. K. Cocke\***, A. Rollett, R. Lebensohn, A. Spear, “The AFRL AM Modeling Challenge: Predicting Micromechanical Fields in AM IN625 Using an FFT-Based Method with Direct Input from a 3D Microstructural Image,” *USNCCM 16*, Virtual, July 2021
1. **C. K. Cocke\***, A. Rollett, R. Lebensohn, A. Spear, “The AFRL AM Modeling Challenge: Predicting Micromechanical Fields in AM IN625 Using an FFT-based Method with Direct Input from a 3D Microstructural Image,” *3DMS 5*, Virtual, June 2021

## Teaching Experience

- Teaching Assistant: Calculus of Variations (ACM 127) Spring 2026
- Teaching Assistant: Continuum Mechanics (ME EN 5530/6530) Fall 2021

## Professional Experience

### **Magnit @ Meta**

June 2025 – Sept. 2025

*Research Scientist III*

Pasadena, CA

- Developed a fully MPI-distributed memory parallelized linear viscoelastic finite-element solver in deal.II for mechanical studies with human tissue geometries

### **Corning**

June 2018 – June 2019

*Engineering Intern*

Salt Lake City, UT

- Fulfilled various engineering tasks to improve automation performance, uptime, and safety through custom designed (SolidWorks), machined (manual/CNC milling), and assembled metal and plastic components

## Honors & Awards

**2023:** National Science Foundation Graduate Research Fellowship (NSF GRFP)

**2022:** Caltech Theodore Y. Wu Graduate Fellowship

**2022:** 1<sup>st</sup> Place: NIST AM-Bench 2022: Subcontinuum Mesoscale Tensile Test

**2022:** Travel Award: 15<sup>th</sup> World Congress on Computational Mechanics (WCCM 15)

**2021:** 1<sup>st</sup> Place: ASME/IEEE Heat Sink Design Competition 2021

**2020:** 1<sup>st</sup> Place: AFRL AM Modeling Challenge Series 2020: Microscale Structure-to-Properties Predictions

**2019:** Larry DeVries Scholarship: awarded to outstanding mechanical engineering students

**2019:** Mechanical Engineering Tuition Waiver

**2017 – 2021:** Utah Flagship Scholarship

## Notable Involvement

**2026:** Mentor: Caltech WAVE fellows program

**2024:** Attendee: Argonne Training Program on Extreme-Scale Computing (ATPESC 2024)

**2024:** Chair: Caltech Knowles Symposium student organizing committee

**2023 – :** Mentor: Caltech Accountability Partners Program (CAPP)

**2023 – :** Officer: Caltech Society of Professional Students (SOPS)

**2022:** Session Chair: 11<sup>th</sup> European Solid Mechanics Conference (ESMC 11)

## Skills

**Languages:** Python, C++, Fortran, MATLAB, C, LaTeX

**Tools:** Git, Linux, Slurm, CMake, MPI, OpenMP, CUDA

**Libraries:** deal.II, FFTW, HDF5, LAPACK

**Software:** DREAM.3D, ParaView, ABAQUS, COMSOL, Gmsh, SolidWorks, VS Code